

Prediction of Video Game Sales Using Machine Learning: Short Project Report

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1 Problem Statement

Predicting video game sales is important for developers and publishers to support marketing strategies, budgeting, and revenue forecasting.

The objective of this project was to develop a machine learning regression system capable of predicting global game sales using historical sales information and engineered features.

2 Methodology

Several preprocessing and feature engineering techniques were applied:

- Missing value handling
- Feature scaling
- Target encoding
- Feature interaction generation
- Regional sales proportion analysis

Regression models used:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Random Forest
- Gradient Boosting
- XGBoost

Model evaluation was performed using:

- MAE
- RMSE
- R^2

3 Results

Model	Test R^2
Linear Regression	0.3475
Random Forest	0.6804
Gradient Boosting	0.7159
XGBoost	0.6740

Gradient Boosting achieved the best performance with a test score of 0.7159.

Tree-based ensemble methods effectively captured non-linear feature interactions and outperformed linear approaches.

4 Conclusion

The project demonstrates that ensemble learning methods provide more accurate prediction of global video game sales than traditional linear models.